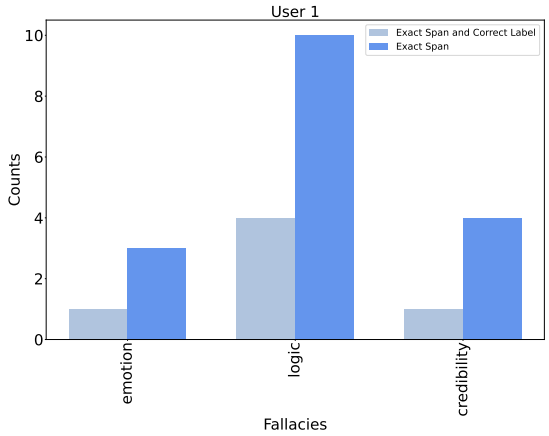
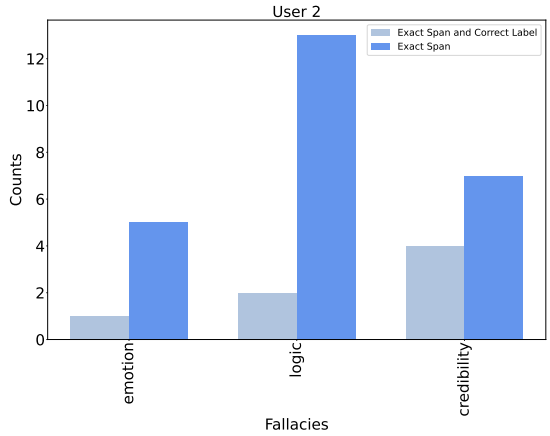
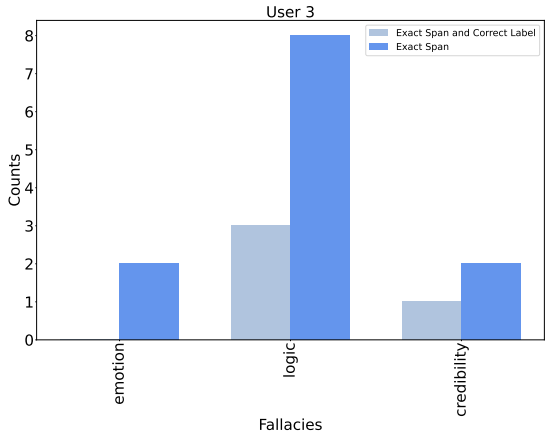


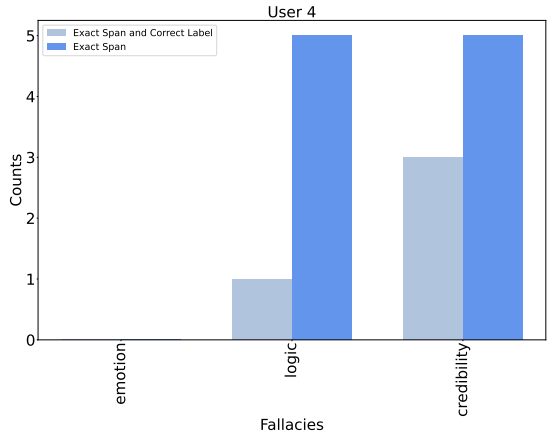
(a) Accuracy of fallacy labeling for spans that exactly match the gold standard at Level 1 for the **User 1**



(b) Accuracy of fallacy labeling for spans that exactly match the gold standard at Level 1 for the **User 2**

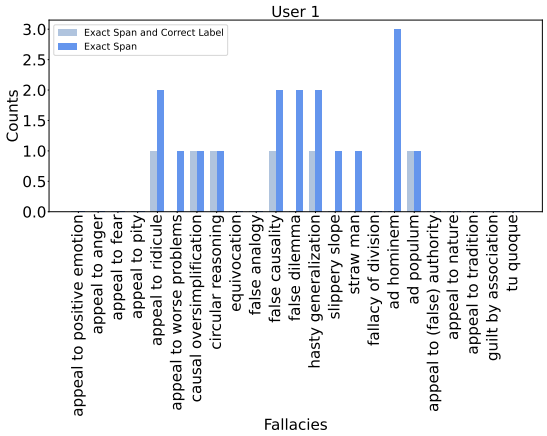


(c) Accuracy of fallacy labeling for spans that exactly match the gold standard at Level 1 for the **User 3**

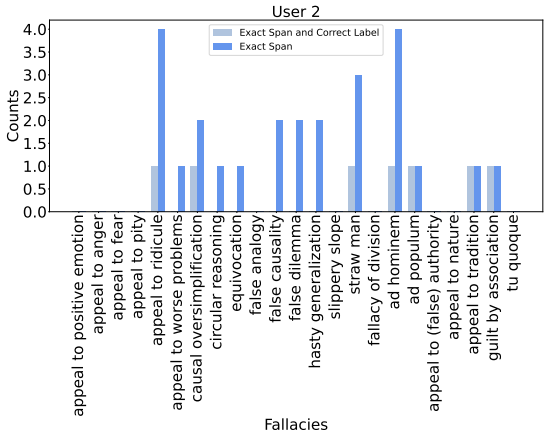


(d) Accuracy of fallacy labeling for spans that exactly match the gold standard at Level 1 for the **User 4**

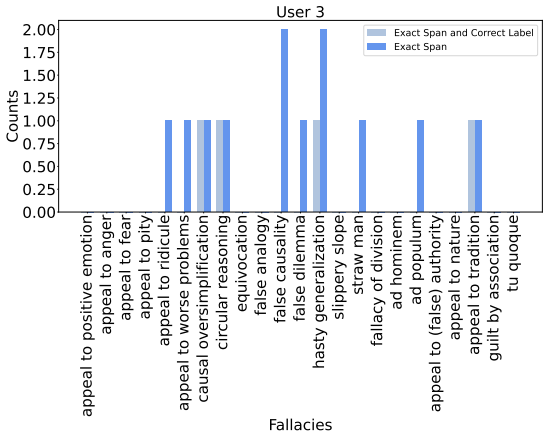
Figure 10: Accuracy of fallacy labeling for spans that **exactly match** the gold standard at Level 1 for the Users' annotations. *Exact Span* corresponds to the number of spans correctly identified by the user, *Exact Span and Correct Label* corresponds to the number of correctly labeled spans out of the correctly identified spans.



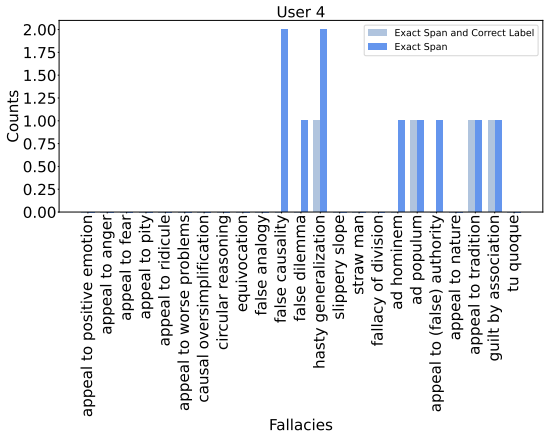
(a) Accuracy of fallacy labeling for spans that exactly match the gold standard at Level 2 for the **User 1**



(b) Accuracy of fallacy labeling for spans that exactly match the gold standard at Level 2 for the **User 2**



(c) Accuracy of fallacy labeling for spans that exactly match the gold standard at Level 2 for the **User 3**



(d) Accuracy of fallacy labeling for spans that exactly match the gold standard at Level 2 for the **User 4**

Figure 11: Accuracy of fallacy labeling for spans that **exactly match** the gold standard at Level 2 for the Users' annotations. *Exact Span* corresponds to the number of spans correctly identified by the user, *Exact Span and Correct Label* corresponds to the number of correctly labeled spans out of the correctly identified spans.

L Edge Cases of the Metrics

In this section, we show how our metrics handle edge cases of our disjunctive annotation scheme.

Table 12: The model predicts at least one correct label

	Spans	Labels		
Gold	Lorem ipsum dolor sit amet.	$l1, \perp$		
	Ut enim ad minim veniam.	$l2$		
	Sed do eiusmod tempor incididunt.	$l3$		
Case	Prediction	Label	Recall	Precision
0.1	Ut enim ad minim veniam.	$l2$	0.5	1
0.2	Lorem ipsum dolor sit amet. Ut enim ad minim veniam.	$l1$ $l2$	0.5	1
0.3	Ut enim ad minim veniam. Sed do eiusmod tempor incididunt.	$l2$ $l4$	0.5	0.5
0.4	Lorem ipsum dolor sit amet. Ut enim ad minim veniam. Sed do eiusmod tempor incididunt.	$l1$ $l2$ $l4$	0.5	0.666

Table 13: The gold standard has only one span, which contains a "no fallacy" as an alternative

	Spans	Labels	Recall	Precision
Gold	Lorem ipsum dolor sit amet.	$l1, \perp$		
Case	Prediction	Label		
1.1	Lorem ipsum dolor sit amet.	$l1$	1	1
1.2	Lorem ipsum dolor sit amet.	$l3$	1	0
1.3	-	-	1	1
1.4	Ut enim ad minim veniam.	$l1$	1	0
1.5	Lorem ipsum dolor sit amet. Ut enim ad minim veniam.	$l3$	1	0

Table 14: The gold standard does not contain a "no fallacy"

	Spans	Labels	Recall	Precision
Gold	Lorem ipsum dolor sit amet.	$l1$		
Case	Prediction	Label		
2.1	Lorem ipsum dolor sit amet.	$l1$	1	1
2.2	Lorem ipsum dolor sit amet.	$l3$	0	0
2.3	Ut enim ad minim veniam.	$l1$	0	0
2.4	-	-	0	1

Table 15: The gold standard contains no fallacious span

	Spans	Labels	Recall	Precision
Gold	-	-		
Case	Prediction	Label		
3.1	Lorem ipsum dolor sit amet.	$l1$	1	0
3.2	-	-	1	1